

HASTELLOY C22 (UNS N06022) – Datasheet

Description:

Hastelloy C22 is a corrosion-resistant nickel-chromium-molybdenum-tungsten alloy, designed to withstand highly aggressive and unpredictable chemical environments. It offers excellent resistance to oxidizing and reducing agents, pitting, crevice corrosion, and stress corrosion cracking. Highly weldable and mechanically strong, C22 is widely used in chemical processing, marine, oil & gas, and critical industrial applications.

Chemical Composition

Element	Composition (%)
Nickel	56 (Balance)
Chromium	22
Molybdenum	13
Iron	3
Tungsten	3
Cobalt	≤2.5
Manganese	≤0.5
Vanadium	≤0.35
Silicon	0.08
Carbon	≤0.01
Phosphorus	≤0.02
Sulfur	≤0.02

Physical Properties

Property	Value
Density	8.89 g/cm ³
Melting Point	1350°C (2460°F)
Tensile Strength	115,000 psi / 790 MPa
Yield Strength	52,000 psi / 355 MPa
Elongation	45%

Mechanical Properties

Property	Metric	Imperial
Tensile Strength (@538°C)	690 – 790 MPa	100,000 – 115,000 psi
Yield Strength (@427°C)	379 MPa	55,000 psi
Elongation at Break (@204°C)	45%	45%
Hardness, Rockwell B (plate)	89 – 100 HRB	89 – 100 HRB

Available Products

- Pipes: Seamless, ERW & EFW
 - Tubes: Capillary, U-Bend
 - Sheets & Plates
 - Coils & Strips
 - Bars: Round, Flat, Hex, Square
 - Pipe Fittings: Elbows, Tees, Reducers, Caps, Stub Ends
 - Flanges: Weld Neck, Slip-On, Blind, Socket Weld, Threaded
 - Fasteners: Bolts, Nuts, Screws, Washers, Studs
 - Custom Fabricated Components
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Key Advantages

- Exceptional corrosion resistance in oxidizing and reducing environments
 - High resistance to pitting, crevice corrosion, and stress corrosion cracking
 - Excellent mechanical properties under chemical and thermal stress
 - Superior weldability and fabrication capability
 - Suitable for aggressive and mixed chemical environments
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Applications

- **Chemical Processing:** Reactors, heat exchangers, scrubbers, storage vessels, piping
 - **Pollution Control / FGD Systems:** Ducts, dampers, scrubbers, stack liners
 - **Oil & Gas:** Sour gas service, subsea equipment, wellhead components, manifolds, separators
 - **Marine & Offshore:** Seawater systems, offshore platforms, marine exhausts
 - **Pulp & Paper:** Bleach plants, digesters, chlorine dioxide generators
 - **Power & Nuclear Plants:** Condensers, pressure vessels, heat exchangers
 - **Pharmaceutical & Food Processing:** Hygienic piping, vessels, systems with high quality standards
 - **Wastewater & Environmental Systems:** Tanks, filters, piping exposed to corrosive effluents
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